

Introduction to Probability and Applied Statistics

Week 2: Describing empirical data

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Parameters and statistics

PARAMETERS

Features of the world. Fixed, unobserved.

Greek letters: μ , σ , β .

STATISTICS

Computed from your data. Observed.

Latin letters: $\bar{X}$, s , b ; hatted, $\hat{\beta}$.

A statistic is to a sample as a parameter is to a population

Estimand, estimator, estimate

The **estimand** θ is the quantity you want, a feature of the population.

The **estimator** $\hat{\theta}$ is a rule that turns a sample into a number.

The **estimate** is that number, for the sample you have.

	Symbol	For the mean age of U.S. adults
Estimand	θ	μ , the mean age of all U.S. adults: fixed, unobserved
Estimator	$\hat{\theta}$	$\bar{X} = \frac{1}{n} \sum_i X_i$: a rule, used to infer the value of a parameter
Estimate	—	53.3: what that rule returns on our 8,000 rows

Estimand, estimator, estimate

ESTIMAND



ESTIMATOR

Ingredients	Directions
150g unsalted butter	1. Heat oven to 160C.
150g chocolate pieces	Grease 1 liter glass
150g all-purpose flour	baking pan. Line a 450g
1/2 tsp baking powder	oaf tin with baking paper.
1/2 tsp baking soda	2. Melt butter and
200g brown sugar	chocolate in a saucepan
2 large eggs	over low heat.

ESTIMATE



Roadmap

Description

The distribution

Location

Dispersion

Covariation

From items to indices

Visualizing Data

Wrapping up

Description is not a lesser mode

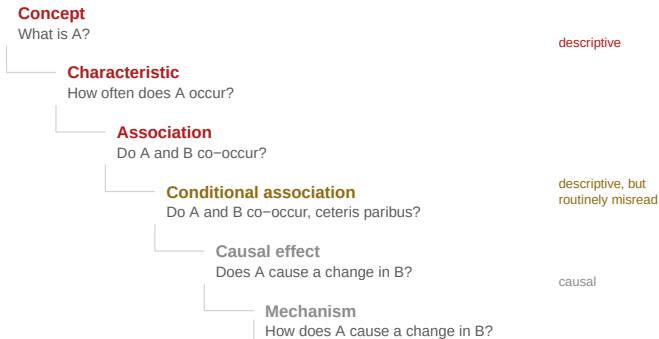
Gerring (2012): description is “derided in favor of causal analysis.”

GRIMMER (2015)

“Much of the empirical work of political scientists and theories that they construct are a direct product of description.”

This section follows de Kadt & Grymala-Busse (2025), “Good Description.”

A ladder of questions



You cannot answer a question on a lower rung without being able to answer the ones above it.

de Kadt & Grymala-Busse (2025), Figure 1.

What makes a question descriptive

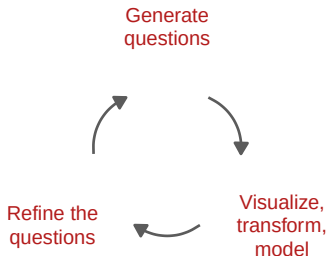
MANIPULABILITY

If answering a question requires that some concept or variable be **manipulable**, the question is causal. If manipulability is not required, the question is **descriptive**.

Race is not manipulable, so *do members of one racial group earn less than others?* is descriptive. *Does this labor-market policy reduce the gap?* is causal.

Holland (1986); de Kadt & Grymala-Busse (2025), section 2.

Exploratory data analysis



Two questions organize the rest of today: what **variation** occurs *within* my variables, and what **covariation** occurs *between* them?

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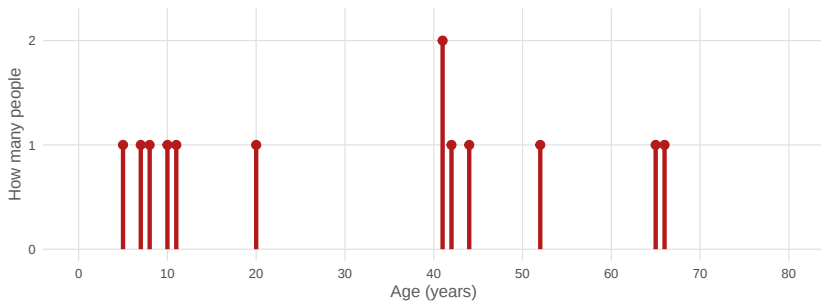
A variable: age

I asked the thirteen people on Westfield Drive how old they are:

5, 7, 8, 10, 11, 20, 41, 41, 42, 44, 52, 65, 66

- What is the average age on the street?
- What is the range of ages?
- What is the age in the middle?
- What is the most common age?

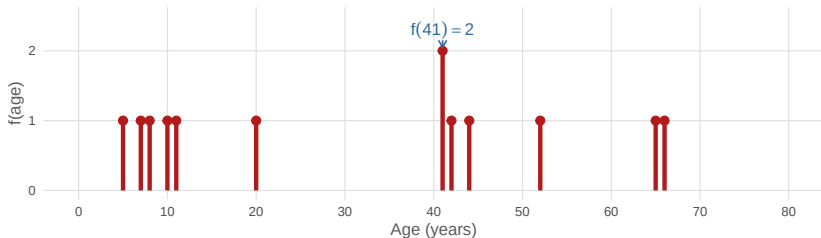
The distribution of age



A distribution is a function

$$f(x) = \sum_{i=1}^n \mathbf{1}\{X_i = x\}$$

Counting exact values only works because age is **discrete** here.



$f(1) = 0$, $f(21) = 0$, $f(41) = 2$, and $\sum_x f(x) = n = 13$.

Divide by n for shares rather than counts: $\hat{p}(x) = f(x)/n$, which sums to 1.

↑
proportion of a sample

Relative frequencies

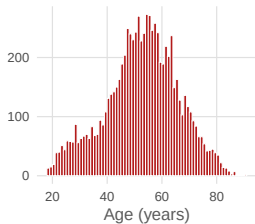
Code	Highest education	Count	Share
1	No HS	166	0.021
2	HS grad	1977	0.248
3	Some college	1874	0.235
4	2-year	784	0.098
5	4-year	1972	0.247
6	Post-grad	1210	0.152

$f(x)$ is the **frequency** of x : the number of observations that take that value.
The Share column is the **relative frequency**,

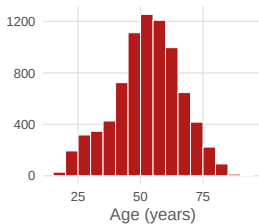
$$\hat{p}(x) = \frac{f(x)}{n} = \frac{\text{number of observations equal to } x}{\text{total number of observations}}.$$

Histograms

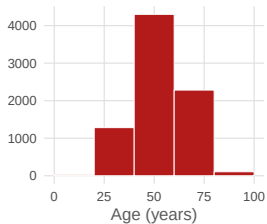
binwidth = 1



binwidth = 5

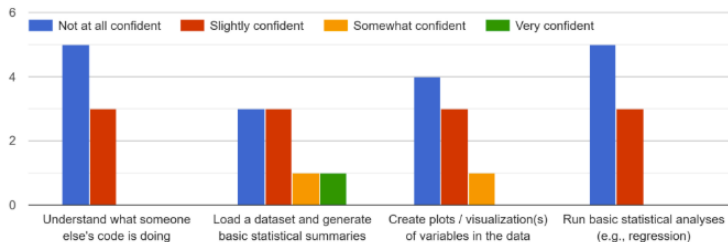


binwidth = 20



...vs. Bar chart

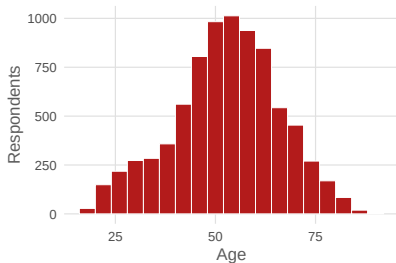
If you have any experience with R, how confident do you feel about your ability to...



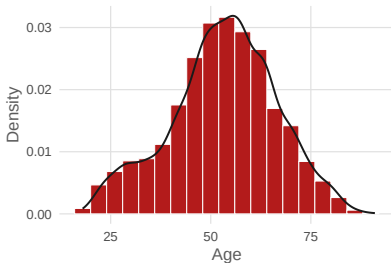
GOVT 6019 background survey, this class.

From counts to density

Count scale



Density scale: total area = 1



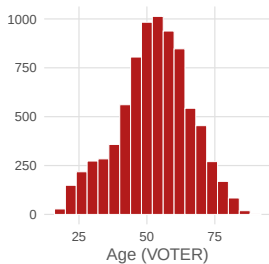
Density is share per unit of x : divide each bar by n and by the binwidth, and bar *area* becomes the share in that bin.

Features of a distribution

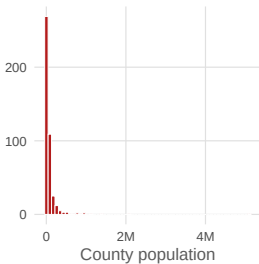
Feature	Named by	Ask
Shape	bell-shaped, bimodal, skewed	Is this what you expected?
Center	mean, median, mode	Which values are common, and why?
Spread	range, IQR, SD	How typical is typical?
Unusual values	outliers	What produced this one?

Shape

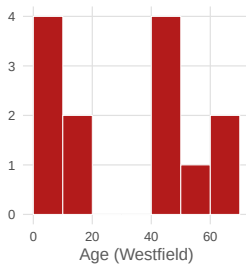
Roughly symmetric



Right-skewed



Bimodal



Roadmap

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Three centers

WESTFIELD DRIVE

5, 7, 8, 10, 11, 20, **41**, 41, 42, 44, 52, 65, 66

- **Mean:** $\bar{X} = \frac{1}{n} \sum_i X_i = 412/13 = 31.69$.
- **Median:** the 7th of 13 sorted values = 41.
- **Mode:** the most frequent value = 41.

Conventions

- If X has an **even** number of elements, the **median** is the mean of the “middle two”.
 - The median of $\{0, 1, 2, 3, 4, 1000\}$ is $(2 + 3)/2 = 2.5$. Two conventions are in use: report that midpoint, which is what software does, or, with Gailmard, treat every value between the middle two as a median.
- The **mode** need not be unique, especially for continuous variables.
 - Each value of $\{-1.2434, 0, 5.6, 9.99, 11\}$ is a mode.
- By definition, the **mean** requires that we can add and divide the values of a variable.
 - What is the mean of $\{\text{Black}, \text{White}, \text{White}, \text{Asian}, \text{White}\}$?
- By definition, the **median** requires that we can order the values of a variable.
 - What is the median of $\{\text{strongly support}, \text{strongly support}, \text{somewhat support}, \text{somewhat oppose}, \text{strongly oppose}\}$?

Order statistics

ORDER STATISTIC

A summary defined by an observation's *position* in the sorted data, not by its value.

Westfield: min = 5, max = 66, **range** = 61, median = 41.

Quantiles

QUANTILE

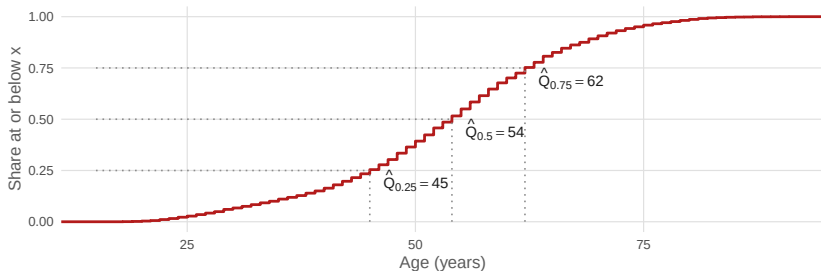
For p between 0 and 1, the p -th **quantile** $\hat{Q}_p$ is the smallest value with at least share p of the data at or below it.

- **Quartiles** cut the data into 4: $\hat{Q}_{0.25}$, $\hat{Q}_{0.5}$, $\hat{Q}_{0.75}$.
- **Quintiles** cut it into 5: $\hat{Q}_{0.2}$, $\hat{Q}_{0.4}$, $\hat{Q}_{0.6}$, $\hat{Q}_{0.8}$.
- **Percentiles** cut it into 100.

Westfield: $\hat{Q}_{0.25} = 10$, $\hat{Q}_{0.5} = 41$, $\hat{Q}_{0.75} = 44$.

The empirical cumulative distribution function

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}$$



eCDF: a step function that jumps by $1/n$ at each of the n data points.

The mean as a weighted sum

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \sum_j x_j \cdot \frac{n_j}{n} = \sum_j x_j \hat{p}(x_j)$$

```
p <- prop.table(table(voter$age)) # share at each distinct age
c(weighted = sum(as.numeric(names(p)) * as.numeric(p)),
  direct   = mean(voter$age))
```

weighted	direct
53.259	53.259

There are two ways to compute the average age of your 8,000 respondents.

1. Add up 8,000 numbers, divide by 8,000.
2. Take each distinct age, multiply it by the fraction of people who have it, and add up those products.

Upshot: the mean is a property of the distribution, not of the rows. The mean is a weighted average of the values, with the relative frequencies as weights.

Weighted means

$$\bar{X}_w = \frac{\sum_i w_i X_i}{\sum_i w_i}$$

A weight of w_i means row i counts as w_i people.
Unweighted is the case $w_i \equiv 1$.

The weighted mean and the unweighted mean are two different estimators.

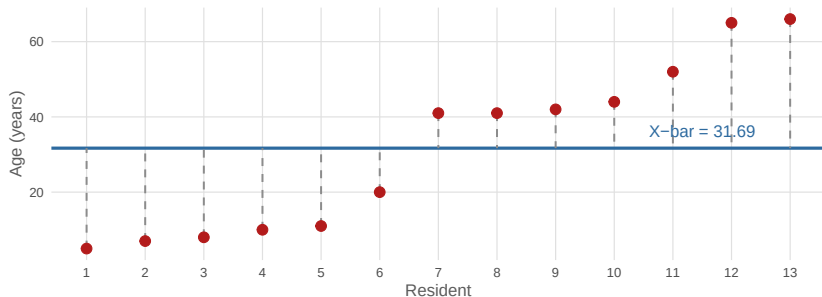
The mean of a 0/1 variable

```
grad <- as.numeric(voter$educ >= 5)      # 1 if 4-year degree or more
c(share = mean(grad, na.rm = TRUE),
  count = sum(grad, na.rm = TRUE))
```

share	count
0.398597	3182.000000

For a variable coded 0/1, the mean **is** the share of 1s (a proportion).

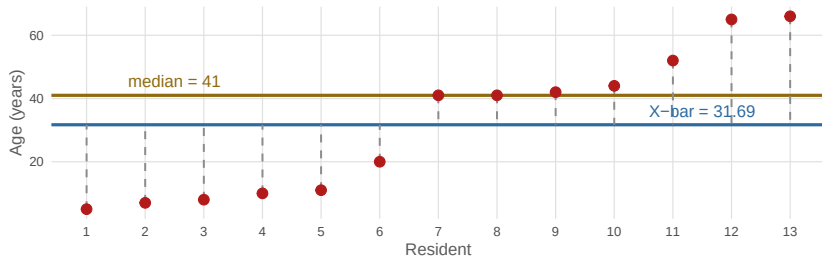
The mean is a prediction



$$\bar{X} = \arg \min_c \sum_i (X_i - c)^2, \text{ uniquely.}$$

The mean is a prediction

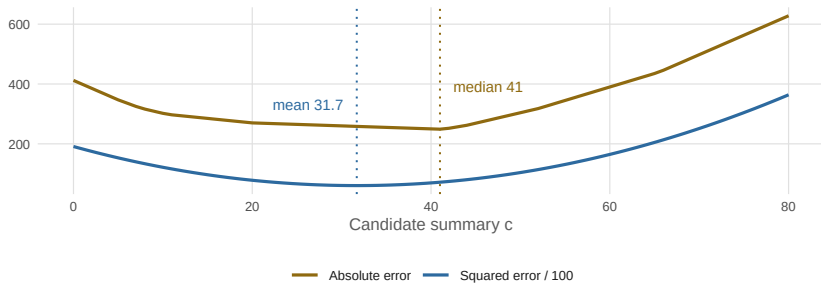
The median is (also) a prediction



$$\bar{X} = \arg \min_c \sum_i (X_i - c)^2, \text{ uniquely.}$$

$$\text{Med}(X) \in \arg \min_c \sum_i |X_i - c|.$$

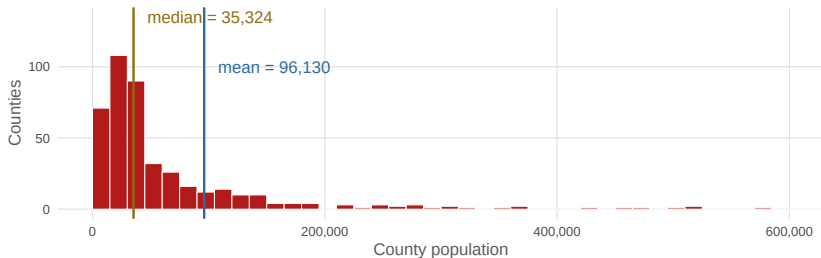
The median minimizes absolute error



$$\text{Med}(X) \in \arg \min_c \sum_i |X_i - c|.$$

Mean and median under skew

What's the population of a 'typical' midwestern county?



10 counties above 600,000 not shown.

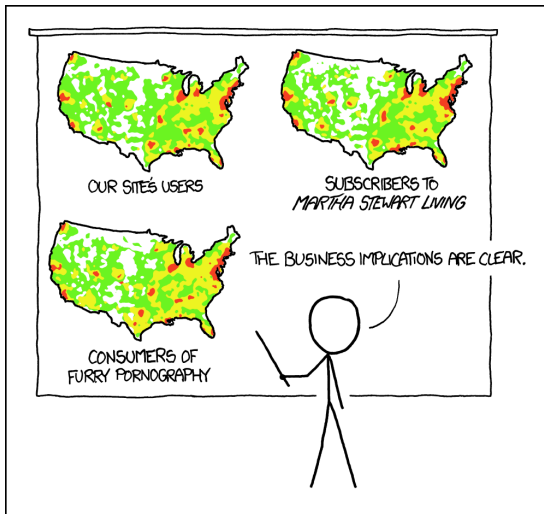
Right skew pulls the mean above the median. Left skew pulls it below, as on Westfield: mean 31.7, median 41.

Denominators and baselines

A count rises. Did the phenomenon spread, or did the denominator grow?

UN membership: **51** states in 1945, **193** today. Any global count has a moving denominator underneath it.

“Is it rising or falling” is not answerable without a **baseline**: a stated reference point in time, and a stated population to be a share of.



PET PEEVE #208:
GEOGRAPHIC PROFILE MAPS WHICH ARE
BASICALLY JUST POPULATION MAPS

xkcd.com/1138

Roadmap

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The distribution

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Dispersion

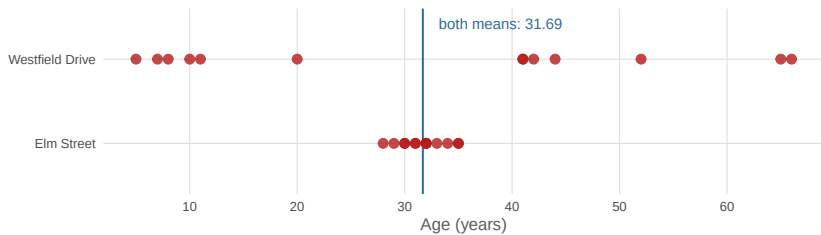
Covariation

From items to indices

Visualizing Data

Wrapping up

Same mean, different data



Range and IQR

INTERQUARTILE RANGE

$\text{IQR} = \hat{Q}_{0.75} - \hat{Q}_{0.25}$: the width of the middle half of the data.

Westfield: range = 61, IQR = 34. Elm: range = 7, IQR = 3.

A value more than $1.5 \times \text{IQR}$ beyond either quartile is conventionally flagged as an **outlier** (but note that there is no singular definition of an 'outlier' or what, if anything, to do about it).

What to do with a flagged observation

- Find out what produced it. A typo, a different unit, a real place.
- Report the summary with it and without it.
- Decide *before* you look at the result, and write the rule down.

Choosing the rule after seeing the result is p-hacking.

Squaring the deviations

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
5	-26.7	712
7	-24.7	610
8	-23.7	561
10	-21.7	471
11	-20.7	428
20	-11.7	137
41	9.3	87
41	9.3	87
42	10.3	106
44	12.3	151
52	20.3	412
65	33.3	1109
66	34.3	1177
412	0.0	6049

Deviations from $\bar{x}$ always sum to zero. Square them first.

The variance

SAMPLE VARIANCE

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Westfield: $s^2 = 6049/12 = 504.1$. Elm: $s^2 = 4.7$.

Why $n - 1$? Because we use the sample mean instead of the true population mean, dividing by n underestimates the variance (Bessel's Correction).

The computational identity

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \overline{X^2} - \bar{X}^2$$

```
x <- voter$age
c(deviations = mean((x - mean(x))^2),
  identity    = mean(x^2) - mean(x)^2,
  reported    = var(x))           # var() uses 1/(n-1)
```

deviations	identity	reported
176.1064	176.1064	176.1284

“Mean of squares minus square of the mean.”

Standard deviation

$s = \sqrt{s^2}$, back on the variable's own scale.

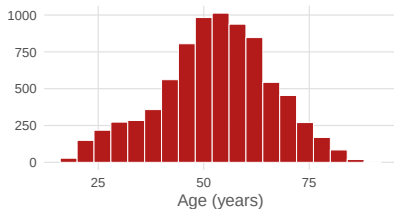
- Only one distinct value in the data $\Rightarrow s = 0$.
- Add k to every observation $\Rightarrow \bar{X}$ moves by k ; s is **unchanged**.
- Multiply every observation by $k \Rightarrow \bar{X}$ is multiplied by k , s by $|k|$.

Westfield $s = 22.5$; Elm $s = 2.2$; VOTER age $s = 13.3$.

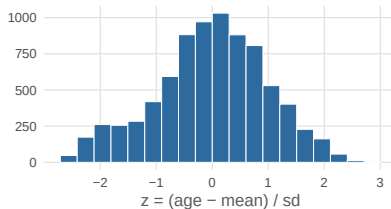
Standardization

$$Z_i = \frac{X_i - \bar{X}}{s} \quad \Rightarrow \quad \bar{Z} = 0, \quad s_Z = 1$$

Age

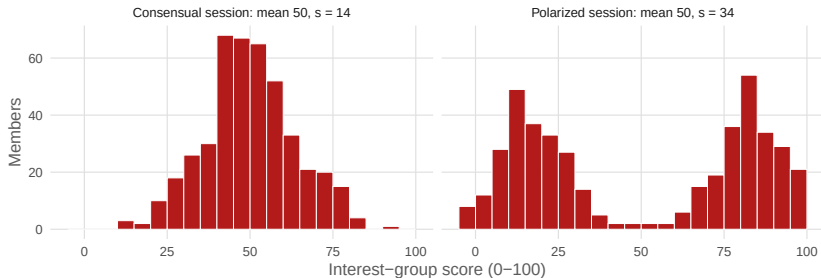


Standardized



Z_i answers: how many standard deviations from the mean?

When spread is the question



The standard deviation of members' scores in a session measures the breadth of the political spectrum in it.

Roadmap

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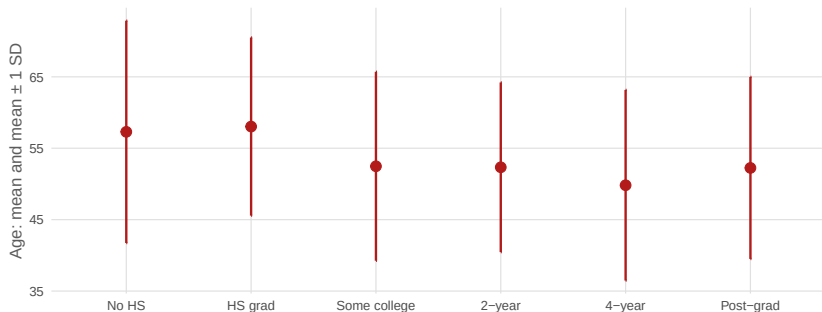
Two variables on the same units

BIVARIATE DISTRIBUTION

An observation of **two** variables for **each** unit in a collection of n units.

Household size for 500 households, appended to income for 500 *different* households, is not a bivariate distribution. It is two univariate distributions in one spreadsheet.

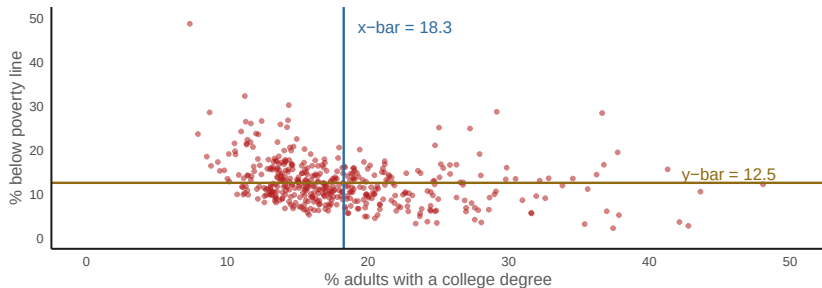
Center and spread by group



Dot = the mean of Y within that group. Bar = $\pm$ the SD of Y within that group.

Scatterplots

College and poverty across 437 midwestern counties



Midwestern counties (IL, IN, MI, OH, WI), 2000. Source: 2000 U.S. Census, via the `midwest` data in the R package `ggplot2`.

Covariance

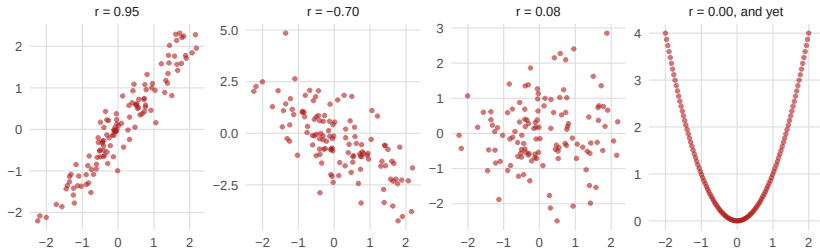
$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_i (x_i - \bar{X})(y_i - \bar{Y})$$

Covariance is a measure of variation between two variables. Sensitive to units (not standardized).

Points on the same side of both means contribute +; opposite sides contribute −.

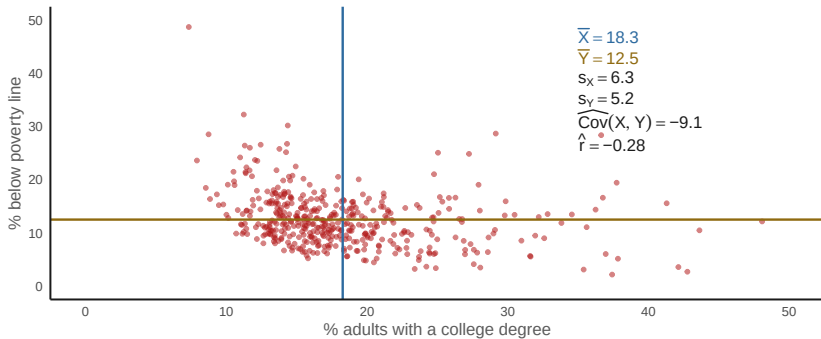
Correlation

$$\hat{r} = \frac{\widehat{\text{Cov}}(X, Y)}{s_X s_Y} \in [-1, 1]$$

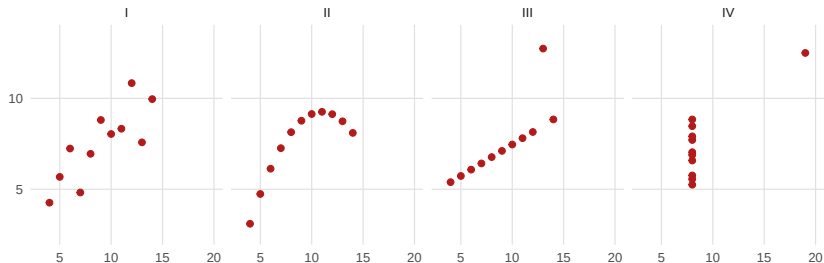


Standardized (unit-free), and thus generally more useful / interpretable than covariance. Reflects *linear* association.

College and poverty across 437 midwestern counties



Anscombe's quartet



All four: $\bar{X} = 9.0$, $\bar{Y} = 7.50$, $s_X = 3.32$, $s_Y = 2.03$, $\hat{r} = 0.82$. Anscombe (1973).

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A model of a survey item

```
n <- 500                                # number of simulated people

# Each person's TRUE importance: a random number between 0 and 1.
# We can see it only because we invented it.
tau <- runif(n)

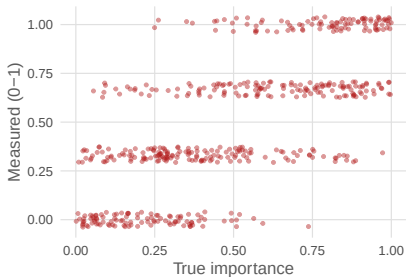
# How a person answers one survey question, given true importance x.
# Answers run 1 = strongly agree ... 4 = strongly disagree.
item <- function(x) {
  answer <- 4 - 3 * x                    # signal: 1 if x = 1, 4 if x = 0
  answer <- answer + rnorm(n, 0, 0.8)    # add random noise
  answer <- round(answer)                # whole-number answers only
  pmin(4, pmax(1, answer))              # keep answers between 1 and 4
}

# A data table, one row per person: tau plus answers to four questions.
# Each item() call draws new noise: same signal, different errors.
sim <- tibble(tau, it1 = item(tau), it2 = item(tau),
              it3 = item(tau), it4 = item(tau))
```

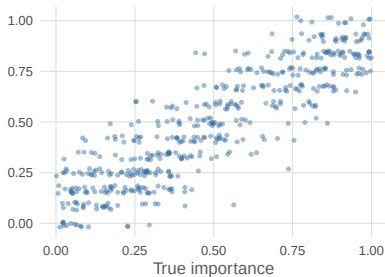
$\text{item}_{ij} = f(\tau_i) + \varepsilon_{ij}$: signal plus noise.

Averaging items recovers the signal

One item: $r = 0.69$



Four-item index: $r = 0.89$



Averaging m items divides the noise SD by $\sqrt{m}$.

Four real items: “importance index”

Item	Wording (agreement, 1–4)	Trap
<code>important_econ</code>	The economy is important	5 = Don't know
<code>important_gayr</code>	Gay rights are important	5 = Don't know
<code>unimportant_budget</code>	The deficit is not important	reverse-framed
<code>unimportant_abort</code>	Abortion is not important	reverse-framed

The scale runs 1 Strongly agree . . . 4 Strongly disagree. **5 = Don't know** is not a fifth level.

Steps 1-2: drop Don't knows, flip

```
dat <- voter_raw |>
  mutate(
    across(c(important_econ, important_gayr,
             unimportant_budget, unimportant_abort),
           \(x) na_if(x, 5)),           # drop Don't knows
    important_budget = 5 - unimportant_budget, # flip 1<->4, 2<->3
    important_abort  = 5 - unimportant_abort   # (after DK removal!)
  )
```

Audit the variables you create

Raw response	new 1	new 2	new 3	new 4	new NA
1 Strongly agree	0	0	0	193	0
2 Agree	0	0	659	0	0
3 Disagree	0	1993	0	0	0
4 Strongly disagree	5027	0	0	0	0
5 Don't know	0	0	0	0	39

Steps 3–4: rescale, average

```
dat <- dat |>
  mutate(across(c(important_econ, important_gayr,
                  important_budget, important_abort),
    \ (x) (4 - x) / 3, # 0 = least, 1 = most important
    .names = "{.col}_01"))

dat$importance <- dat |>
  select(ends_with("_01")) |>
  rowMeans(na.rm = TRUE)
```

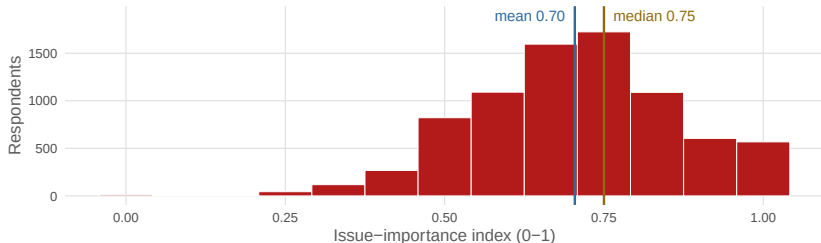
$(4 - x)/3$ maps $4 \mapsto 1$ and $1 \mapsto 0$: higher is more important, which matches the name.

The finished index

```
dat |>
  summarize(scored = sum(!is.na(importance)),
            mean   = mean(importance, na.rm = TRUE),
            median = median(importance, na.rm = TRUE),
            sd     = sd(importance, na.rm = TRUE))
```

A tibble: 1 x 4

	scored	mean	median	sd
	<int>	<dbl>	<dbl>	<dbl>
1	7949	0.704	0.75	0.162



Is this index any good?

Last week's question	What today's tools answer it with
Reliable?	Do the four items vary together? (see Cronbach's α , built from variances)
Valid?	Does the index correlate with other variables in the way you'd expect (e.g., political interest)?

Economy, gay rights, deficit, abortion. One attitude, or four?

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What is Visualization?

Visualization: Presenting something visually (wow, insightful).

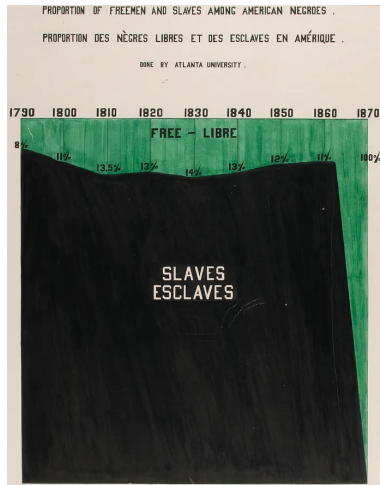
This could mean a:

- histogram or density plot of a variable
- two-way (or three-way) scatterplot of points
- map of something
- photo or screenshot of an experimental intervention
- coefficient plot
- trend-line over time
- table (!)

Visualization is important through the life-cycle of a project:

- **Early:** looking at data to assess quality, properties, face validity
- **Mid:** monitoring what you are doing along the way
- **Late:** what you include in presentations/papers

Good Visualization, Bad Visualization

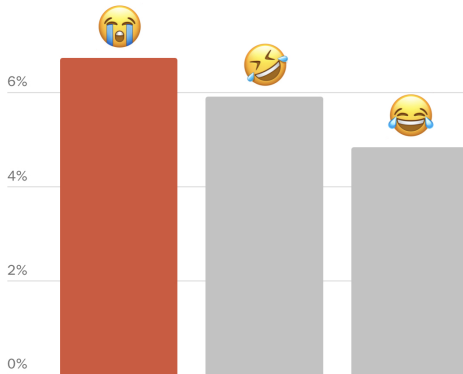


W. E. B. Du Bois and Atlanta University, "Proportion of freemen and slaves among American Negroes," Paris Exposition, 1900. Library of Congress.

Good Visualization, Bad Visualization

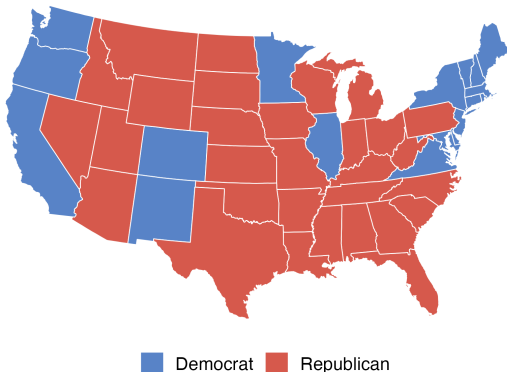
The most popular emojis in 2025

8% of all emoji used



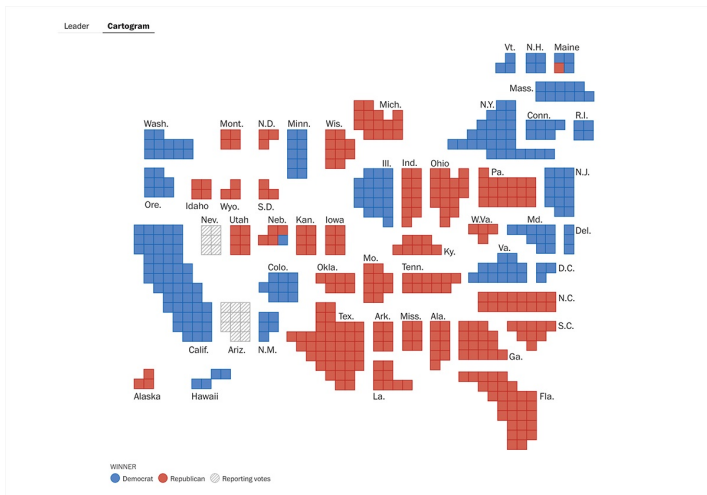
Eve Washington, Larry Buchanan & Claire Cain Miller, *The New York Times* (The Upshot), August 12, 2026. Data: Google Gboard.

Good Visualization, Bad Visualization



2024 U.S. presidential election, winner by state. Alaska (R) and Hawaii (D) not shown.
Results: Federal Election Commission.

Good Visualization, Bad Visualization



2024 U.S. presidential election. *The Washington Post*, November 2024.

The Language of Visualization: Some Anatomy

A visualization is some visual representation drawn on a plane, with four necessary components.

- **Data:** Representation of the information we care about
- **Aesthetics:** Symbolic features, e.g.
 - Coordinate [x,y]
 - Color
 - Shape
- **Mappings:** Correspondence between data and aesthetics
- **Geometries:** Objects that are drawn on the plane, e.g.
 - Lines
 - Points
 - Bars

A visualization is a **mapping** of **data** to **aesthetics** via visual **geometries** to produce meaning. Formally this can be represented as a “**grammar of graphics**” (Wilkinson, 2005).

The Language of Visualization: Some Anatomy

Mappings of data to aesthetics and geometries give visualizations **meaning** (or 'content'). Note what, strictly speaking, doesn't matter...

- **Explanations:** A language-based statement of what the plot represents
 - A note at the bottom of the plot
 - Titles and subtitles
 - Axis labels
 - Legends
- **Design:** Things you do to make the plot look good
 - Typeface
 - Font
 - Themes

Of course this stuff **does matter**, but not in the way we imagine. Let's return to it later.

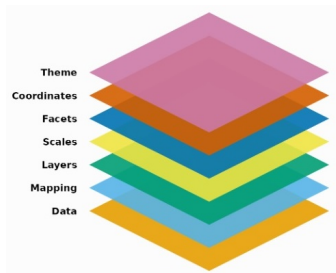
Grammar of Graphics $\longleftrightarrow$ {ggplot2}

In {ggplot2}, three key pieces:

1. **Data:** The information
2. **Mapping:** Correspondence between data and aesthetics
3. **Layers:** Geometry and transformation

Then we get:

- Scales: Paired to an aesthetic
- Facets: Subsets of data
- Coordinates: The x-y system of the canvas
- Theme: Design choices



Source: Introduction to ggplot2

The Good, The Bad, and the Ugly

Misconception: Good data visualization = good visual taste.

No! Good (and bad) data visualization is **not** primarily about how nice (ugly) your plot looks.

It is about **communication** with the **reader/viewer**.

Making your plot look nice is the **last step** in data/model visualization, and only relevant once your visualization meets its “communication goal.”

If your visualization is misleading, confusing, or fails to communicate, then from a scientific perspective it **does not matter** how pretty it is.

Design and explanation only matter inasmuch as they contribute to the communication goal.

Goals of Visualization

What are useful **communication goals**?

- Delivering a (set of) fact(s)
- Telling a story
- Imparting meaning

So, what makes for a good (bad) visualization:

- **Thoughtful**: There is a clear communication goal, and the visualization is built with that goal in mind.
- **Honest**: The reader is not misled about the data or the meaning.

Good vs. Bad: General Principles

Emphasis: **thoughtful** and **honest** visualization.

Thoughtful:

- Communication goal you can articulate in words
- One aesthetic mapping per variable (some exceptions apply)
- No unmapped aesthetics
- Visually interpretable mappings
- Balance information and meaning – raw information is rarely meaningful
- Avoid clutter (“chartjunk” in Tufte’s (1983) words)

Honest: - Avoid distortions of the data or statistics - Avoid axis compression where possible/necessary - Avoid “made up” data

Good vs. Bad: The Lie Factor

Tufte (1983) proposes the Lie Factor:

$$\text{Lie Factor} = \frac{\text{Size of Comparison in Visual}}{\text{Size of Comparison in Data}}$$

Where:

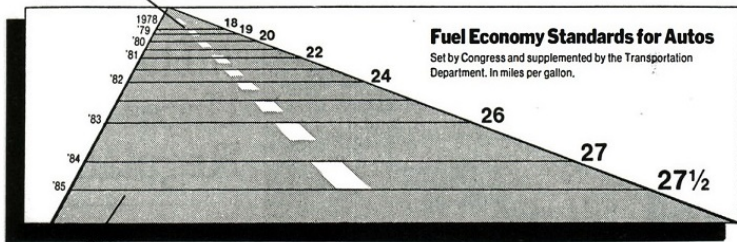
$$\text{Size of Comparison} = \frac{\text{Second Value} - \text{First Value}}{\text{First Value}}$$

Essentially, any Lie Factor other than 1 implies a visual distortion of the data.

Good vs. Bad: The Lie Factor

Classic case from the New York Times:

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.

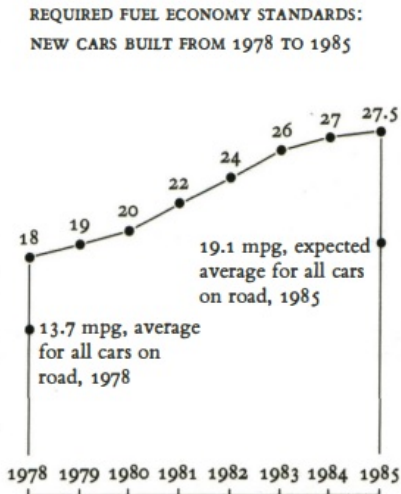


This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.

Tufte calculates the Lie Factor of this as $14.8 = \frac{783\%}{53\%}$

Good vs. Bad: The Lie Factor

Undoing the lie, mostly (Tufte, 1983):



Roadmap

Description

The distribution

Location

Dispersion

Covariation

From items to indices

Visualizing Data

Wrapping up

Further reading

- Gailmard, sections 2.2–2.3.
- de Kadt & Grymala-Busse (2025), “Good Description.”
- Aronow & Miller, sections 1.1–1.2.
- Healy, *Data Visualization*, ch. 1: <https://socviz.co/>.
- Ansolabehere, Rodden & Snyder (2008), “The Strength of Issues,” *APSR* 102(2).
- Little & Meng (2024), “Measuring Democratic Backsliding,” *PS* 57(2), 149–161.
- Choi (2022), “Outlier analysis: Natural resources and immigration policy,” *PLoS One* 17(1).